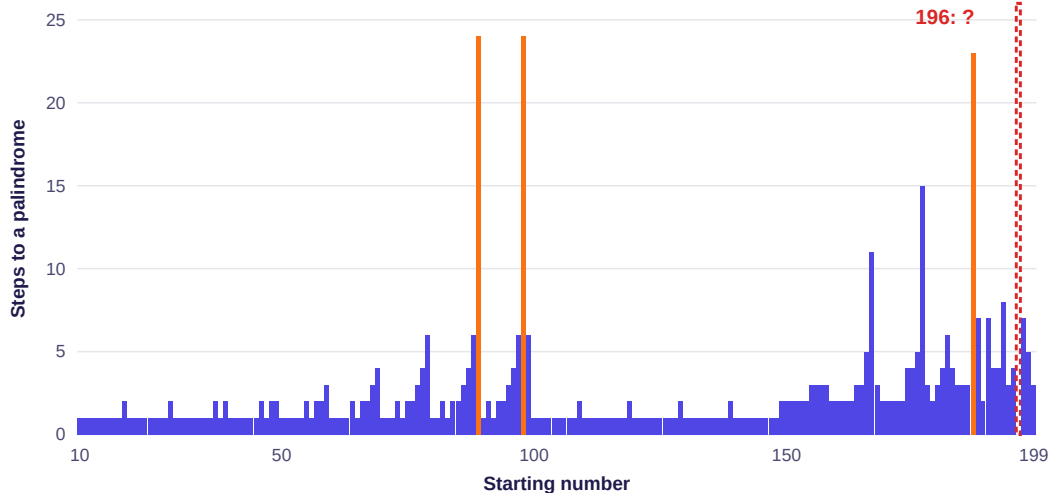


The 196 Problem

Reverse a number and add. Repeat until it reads the same both ways. Does 196 ever get there?



How many steps each starting number from 10 to 199 needs. Most need one or two. 196 has never finished.

TRY IT

$$57 + 75 = 132$$

$$132 + 231 = 363 \text{ (done in 2 steps)}$$

$$89 \text{ takes 24 steps, ending at } 8,813,200,023,188$$

$$196 \rightarrow 887 \rightarrow 1,675 \rightarrow 7,436 \rightarrow \dots$$

WHAT MATHEMATICIANS KNOW

- Computers have followed 196 for hundreds of millions of digits without finding a palindrome.
- A number that never reaches a palindrome is called a Lychrel number. No one has proved that one exists in our base-ten system.
- In some other number systems, like binary, certain numbers have been proved to never reach a palindrome.

WHY IT'S STILL UNSOLVED

Each step makes the number longer, and carrying scrambles the digits in hard-to-predict ways. There's no known way to know what happens at step one billion without doing the first billion steps.

TRY THIS WITH STUDENTS

- Give each group a range of starting numbers (10–29, 30–49...). Pool the results: which are fast, which are slow, which seem stuck?
- Ask: Which two-digit numbers finish in one step? Why those? (Add the two digits.)
- Ask: 89 and 98 take the same number of steps. Why?

WORDS TO KNOW

Palindrome a number that reads the same both ways, like 363

Reverse write the digits in the opposite order